



Neoadjuvant and Adjuvant Systemic Therapy for Early-Stage Non-small-Cell Lung Cancer

James Isaacs¹ · Thomas E. Stinchcombe¹

Accepted: 20 April 2022 / Published online: 21 May 2022
© The Author(s), under exclusive licence to Springer Nature Switzerland AG 2022

Abstract

Approximately a third of patients with non-small-cell lung cancer (NSCLC) present with surgically resectable disease. Patients who undergo surgical resection are at a high risk of relapse, and neoadjuvant and adjuvant chemotherapy improves disease-free survival (DFS) and overall survival (OS). The outcomes with neoadjuvant and adjuvant chemotherapy are similar, and both are used in clinical practice. Recent trials investigated the role of immunotherapy and targeted therapy in patients with early-stage disease. A phase III trial of adjuvant atezolizumab compared with standard of care (SOC) in patients with resected stage II or III disease and PD-L1 expression of 1% or greater, and a second trial of adjuvant pembrolizumab compared with placebo in patients with stage IB–III (regardless of tumor proportion score PD-L1 expression), both demonstrated an improvement in DFS. In the neoadjuvant setting, results of a phase III trial of chemotherapy and nivolumab compared with chemotherapy alone revealed an improvement in pathological complete response rate and event-free survival in patients with stage IB–IIIA disease. Finally, for epidermal growth factor receptor (*EGFR*) mutant NSCLC, a phase III trial of osimertinib compared with SOC revealed an improvement in DFS. The results of these and ongoing trials illustrate the integration of immunotherapy and targeted therapies into the treatment paradigm of patients with surgically resected NSCLC and have led to FDA and EMA approvals in selected populations. Neoadjuvant trials have investigated novel endpoints such as major and complete pathological response, which have the potential to serve as surrogate endpoints for future trials.

Key Points

Adjuvant immunotherapy with atezolizumab or pembrolizumab are treatment options for patients with resected non-small-cell lung cancer (NSCLC).

Neoadjuvant chemotherapy with nivolumab is a treatment option for patients with resectable NSCLC.

Adjuvant osimertinib is a treatment option for patients with resected NSCLC and a sensitizing epidermal growth factor receptor mutation.

1 Introduction

Surgical resection of non-small-cell lung cancer (NSCLC) is potentially curative for patients with stage I and II disease and select patients with stage IIIA disease [1]. However, even after complete surgical resection, a significant proportion of patients experience local and/or distant disease recurrence due to occult micrometastatic disease at the time surgical resection. Several phase III trials investigated surgical resection alone compared with surgical resection with adjuvant cisplatin-based chemotherapy. In a pooled analysis of trials of adjuvant cisplatin-based therapy, the addition of cisplatin-based chemotherapy improved overall survival (OS) (hazard ratio [HR] of 0.89, 95% confidence interval [CI] 0.82–0.96; $p = 0.005$) with an improvement in the 5-year OS rate of 5.4% [2]. The benefit was observed in patients with stage II and III disease. Additional trials investigated the role of neoadjuvant cisplatin-based chemotherapy, and a meta-analysis of neoadjuvant chemotherapy, revealed a similar improvement in OS (HR 0.84, 95% CI 0.77–0.92; $p = 0.0001$) [3]. Thus, adjuvant and neoadjuvant chemotherapy are considered standard treatment options.

✉ James Isaacs
james.isaacs@duke.edu

¹ Division of Medical Oncology, Duke Cancer Institute, Duke University Medical Center, DUMC 3841, Durham, NC 27710, USA

The treatment approach used depends on the individual patient, the stage, patient preferences, and local practice patterns. There is widespread recognition that the efficacy of adjuvant and neoadjuvant cisplatin-based chemotherapy is limited, and there is a clinical need for more efficacious therapies for patients with early-stage NSCLC.

Historically, therapeutic advances in the metastatic setting have been applied to the adjuvant setting. Bevacizumab, an anti-angiogenesis monoclonal antibody, and gefitinib and erlotinib, epidermal growth factor receptor (EGFR) tyrosine kinase inhibitors (TKIs), demonstrated activity in patients with metastatic disease. Bevacizumab had demonstrated an improvement in OS in patients with metastatic non-squamous NSCLC, but a phase III trial that compared chemotherapy alone or with bevacizumab did not reveal an improvement in OS in resected NSCLC [4]. Similarly, a phase III trial of erlotinib compared with placebo, and a phase III trial of gefitinib compared with placebo in resected NSCLC did not reveal an improvement in OS [5, 6]. Importantly, patients were enrolled in the erlotinib and gefitinib trials without regard to *EGFR* mutation status. Thus, the initial attempts to incorporate targeted or biological therapies into surgically resected disease were met with difficulties.

The recent development of immune checkpoint inhibitors alone or in combination with chemotherapy for metastatic disease, and the activity of EGFR TKIs in patients with a sensitizing *EGFR* mutation, have renewed interest in developing these agents in surgically resected patients. Trials have investigated both neoadjuvant and adjuvant strategies, and most of the neoadjuvant trials include adjuvant therapy as well. Several trials have led to the approval of systemic therapies in surgically resected NSCLC, and we anticipate the results of ongoing trials to be available within the next year.

2 Adjuvant Therapy: Immune Checkpoint Inhibitors

Atezolizumab is the first immune checkpoint inhibitor to receive United States (US) Food and Drug Administration (FDA) approval as an adjuvant therapy for PD-L1 expressing stage II to IIIA NSCLC (defined as PD-L1 expression of $\geq 1\%$) following surgical resection and completion of adjuvant cisplatin-based chemotherapy. PD-L1 expression was assessed using the SP263 immunohistochemistry (IHC) assay. The approval is based on the IMpower010 trial where atezolizumab demonstrated a disease-free survival (DFS) benefit in this population in comparison with observation alone following adjuvant chemotherapy (HR 0.66, 95% CI 0.50–0.88; $p = 0.0039$) [7]. The benefit appeared to be most pronounced in patients with PD-L1 expression $> 50\%$. Further, patients who had *EGFR* or *ALK* genomic alterations, or had no prior smoking history,

appeared to have less benefit. As these analyses were driven by exploratory subgroups, there remains substantial debate over which patient populations should be selected for adjuvant checkpoint inhibitors.

Additionally, an interim analysis of a trial evaluating adjuvant pembrolizumab has recently been reported [8]. The PEARLS trial evaluated adjuvant pembrolizumab compared with placebo in patients with IB to IIIA NSCLC following surgical resection and chemotherapy (which was strongly recommended for stage II and II and considered for Stage IB disease, but not required). PD-L1 was determined using the PD-L1 IHC 22C3 assay, and the tumor proportion score (TPS). This trial met its primary endpoint prolonging DFS in the entire population (HR 0.76, 95% CI 0.63–0.91). Surprisingly, the dual primary endpoint of DFS in the TPS PD-L1 $\geq 50\%$ was not met (HR 0.82, 95% CI 0.57–1.18). An exploratory subgroup analysis suggested that the DFS benefit may have been driven by the TPS PD-L1 1–49% group [8].

Overall survival data are awaited from IMpower010 and PEARLS. There are two additional pivotal phase III trials evaluating durvalumab (ClinicalTrials.gov identifier: NCT02273375) and nivolumab (NCT02595944) [9]. Additionally, biomarkers to select patients most likely to benefit from checkpoint inhibitors in the early-stage setting are being pursued, with trials powered to detect DFS in patients with molecular residual disease (detectable circulating tumor DNA) (NCT04385368, NCT04642469).

3 Neoadjuvant Therapy

Advantages of neoadjuvant therapy include better tolerance of therapy prior to surgery, earlier treatment of micrometastatic disease, and the ability to assess treatment response at the time of surgical resection. In some cases, neoadjuvant treatment may also downstage tumors to allow for complete surgical resection. However, these potential benefits must be weighed against the risks of treatment toxicity or disease progression delaying or preventing a potential surgical cure [10, 11]. The current era of immune therapy has ushered in a renewed interest in neoadjuvant therapy given the theoretical advantage of enhanced anti-tumor immune activation in comparison with adjuvant immune therapy (as discussed below in section 3.2) [12]. Additionally, whereas adjuvant trials can take 5–10 years to complete, trials evaluating neoadjuvant therapy provide an early endpoint with the pathologic assessment of tumor response in the surgical resection specimen. Thus, a major focus of ongoing research efforts will be to validate pathologic endpoints and determine which combinations of immune therapies offer the most potent anti-tumor immune activity in the neoadjuvant setting.

3.1 Role of Pathologic Response

Early retrospective studies of patients with NSCLC treated with neoadjuvant chemotherapy suggested that a complete pathologic response or absence of viable tumor at the time of surgical resection correlated with favorable long-term survival, a finding validated in several prospective trials [13, 14]. However, complete pathologic response is a rare finding with chemotherapy alone, with a reported occurrence of < 5% of patients across multiple neoadjuvant chemotherapy trials [15]. Recognizing this limitation, Junker et al. originally validated a histopathologic scoring system suggesting that a cut off of 10% or less of viable tumor correlated with long-term survival [16]. Because the determination of the percentage of viable tumor cells is subject to pathologic interpretation, the International Association for the Study of Lung Cancer (IASLC) has attempted to standardize this process by publishing a formal recommendation. A major pathologic response (MPR) is defined as 10% or less of viable tumor in comparison to necrosis and stromal cells (immune cells and fibrosis) in the tumor bed [17]. This threshold for MPR has now been confirmed to predict OS in both large retrospective cohorts and prospective trials [18, 19]. However, the use of a universal 10% threshold for MPR may not be straightforward as an analysis has suggested, and the optimal cut off to predict OS may differ between squamous cell and adenocarcinoma [20]. Despite this limitation, these studies form a basis for the use of MPR or pathologic complete response (pCR) as an endpoint for trials evaluating neoadjuvant immune therapy. Further data are needed to correlate pathologic responses with long-term outcomes such as DFS or OS, and determine the optimal cut off for pathologic response in the immune checkpoint inhibitor setting. New scoring systems, such as the ‘immune-related pathologic response criteria’ are also being developed to characterize pathologic features specific to checkpoint inhibitor treatment response.

3.2 Features of Immune Response

Initial studies evaluating neoadjuvant checkpoint inhibitors show unique changes suggestive of immune activation in the tumor microenvironment. Termed immune-related pathologic response criteria (irPRC), these features include immune activation (presence of tumor infiltrating lymphocytes and tertiary lymphoid structures), cell death (cholesterol clefts), and tissue repair (neovascularization and proliferative fibrosis) [21]. This response is seen in NSCLC as well as several other tumor types including melanoma and renal cell carcinoma [22]. IrPRC differs from standard pathologic response criteria by defining a regression bed which includes the features described above and often surrounds the periphery of residual tumors that

have responded to immune checkpoint blockade. In this case, the regression bed is included in the total tumor bed area denominator of the equation to determine percentage residual viable tumor and thus may affect the estimated percentage response (viable tumor/total tumor bed area) [22]. By defining the characteristics of an immune-mediated regression bed, initial studies have suggested higher reproducibility between pathologists [22]. Since the irPRC is a new concept, the recent IASLC guideline does not recommend the use of irPRC until further validation in prospective studies is performed [17].

In addition to clear immune activation in the tumor microenvironment, neoadjuvant therapy also contributes to systemic anti-tumor immunity. As there is a greater presence of tumor antigen with the tumor in situ, neoadjuvant immune checkpoint inhibition may generate enhanced activity of tumor-antigen-specific T cells in comparison with adjuvant therapy. This principal has been shown in a murine model with superior tumor control after neoadjuvant checkpoint inhibition rather than adjuvant therapy, mediated by a greater portion of neoantigen-specific T cells seen circulating in the peripheral blood [23]. Similar findings of enhanced peripheral expansion of neoantigen-specific T cells was seen in the neoadjuvant setting in a clinical trial randomizing patients to neoadjuvant or adjuvant ipilimumab + nivolumab as treatment for melanoma [24].

4 Neoadjuvant Immune Checkpoint Inhibition in Non-small-Cell Lung Cancer

The first published trial of neoadjuvant immune checkpoint inhibition in NSCLC evaluated two cycles of nivolumab prior to surgical resection (Table 1). In this trial, there was an MPR in 9 of 20 resected tumors (45%), with a pCR in 2 (10%) [25]. Correlative analysis showed rapid expansion of neoantigen-specific T-cell clones in the peripheral blood, which were also seen to expand in tumor bed. Following this, the larger LCMC3 trial evaluating neoadjuvant atezolizumab showed a more modest MPR rate of 21% and 7% pCR [26]. Several other trials have investigated different immune checkpoint inhibitors, and the results were similar (Table 1) [25–30]. Importantly, in these trials neoadjuvant anti-PD-1/L1 therapy did not adversely impact surgical outcomes. Most of these trials were single-arm trials with safety, pathology or feasibility endpoints, and enrolled patients with a variety of stages. PD-L1 expression was not an eligibility criteria. These trials have successfully demonstrated the safety of this approach, but the single-arm design used limits interpretation of the efficacy results.

Table 1 Select clinical trials of pre-operative immune checkpoint inhibitors

Study	No. of patients	Stage	Therapy	Primary endpoint	ORR (%)	MPR (%)
Forde et al. [25]	22	I–IIIA	Nivolumab	Safety and feasibility	10	45
Lee et al. [26]	147	IB–IIIB	Atezolizumab	MPR	7	20
Cascone et al. [27]	44	I–IIIA	Nivolumab Nivolumab/ipilimumab	MPR	22 19	22 38
Gao et al. [28]	40	IA–IIIB	Sintilimab	Safety	20	40.5
Wislez et al. [29]	50	IB–IIIA	Durvalumab	R0 surgical resection	9	19
Besse et al. [30]	50	I–IIIA	Atezolizumab	Safety	7	14

MPR major pathological response rate, ORR objective response rate

4.1 Immunotherapy Combination Regimens

After initial safety was demonstrated with anti-PD-1 monotherapy, additional trials have examined combination regimens in the neoadjuvant setting. A single-center randomized trial demonstrated superior pathologic response rates with neoadjuvant ipilimumab and nivolumab (38% MPR, 29% pCR) in comparison with nivolumab monotherapy (22% MPR, 9% pCR) [27]. This trial also showed greater immune activation with ipilimumab + nivolumab as there was greater expansion of effector memory and tissue-resident T cells in the tumor microenvironment. Surgical outcomes did not appear to be compromised by the ipilimumab and nivolumab regimen. In contrast, a separate trial evaluating neoadjuvant ipilimumab and nivolumab closed after nine enrolled patients due to high rates of treatment-related adverse events and disease progression, which precluded surgery [31]. Thus, despite greater immune-mediated responses, the addition of ipilimumab does carry significant risk of toxicity and a significant portion of patients will still have tumors that remain nonresponsive to immune therapy approaches.

4.2 Chemotherapy and Immune Checkpoint Inhibitors

Chemotherapy may offer more predictable disease control and toxicity profile than ipilimumab. The combination of chemotherapy and immunotherapy is standard for metastatic patients, and efficacy is observed in patients with low, moderate, and high PD-L1 expression. Thus, the majority of recently published phase II trials (Table 2)

[32–34] and ongoing phase III trials are evaluating combinations of anti-PD-1/L1 therapies with chemotherapy. The phase III Checkmate 816 trial recently reported interim efficacy data, and 358 patients were randomized to three cycles of nivolumab and platinum doublet chemotherapy compared with chemotherapy alone. The trial revealed statistically significant higher pathologic responses with nivolumab and chemotherapy compared with the chemotherapy alone arm (24% and 2.2%, respectively; odds ratio 13.94, 99% CI 3.49–55.75; $p < 0.0001$) [35]. Benefit from the addition of nivolumab was seen regardless of PD-L1 status (assessed using the 28-8 Immunohistochemistry), though numerically higher responses were seen in PD-L1 expressing tumors in comparison with tumors not expressing PD-L1 in the nivolumab arm. Importantly, 83% of patients received definitive surgery in the nivolumab arm versus 75% in the chemotherapy arm, with surgery cancelled due to adverse events in only two patients in each arm. A statistically significant longer event-free survival was observed in the nivolumab-containing arm compared with the chemotherapy arm (HR 0.63, 97.38% CI 0.43–0.91; $p = 0.005$). The final analysis of the association between pathological response and event-free survival is eagerly awaited, as initial smaller phase II trials have suggested that for anti-PD-1/L1 therapies a pCR or MPR is predictive of DFS and OS [33, 34]. Ultimately, the efficacy and favorable safety data with limited impact on surgical resection completion in Checkmate 816 may lead to a substantial shift in the treatment paradigm for patients with early-stage disease.

Table 2 Select phase II trials of chemotherapy and immune checkpoint inhibitors

Study	No. of patients	Stage	Therapy	Primary endpoint	ORR (%)	MPR (%)
Shu et al. [32]	30	IB–IIIA	Atezolizumab + carboplatin/paclitaxel	MPR	63	57
Provencio et al. [33]	46	IIIA	Nivolumab + carboplatin/paclitaxel	PFS at 24 mo	76	83
Rothschild et al. [34]	67	IIIA	Cisplatin/docetaxel durvalumab	EFS at 12 mo	58	62

EFS event-free survival, MPR major pathological response, ORR objective response rate, PFS progression-free survival

4.3 Ongoing Questions for Immune Checkpoint Inhibitors

Initial clinical trials suggest a benefit of immune checkpoint inhibitors in either the adjuvant or neoadjuvant setting. While there are multiple ongoing trials in both settings (Table 3), there are no direct comparisons between the two approaches. It is likely that in the future both neoadjuvant and adjuvant combinations of immune checkpoint inhibitors will be an accepted standard of care. Whether certain populations would benefit more from a specific approach will remain a topic of debate. For example, in IMpower010, PD-L1 negative tumors did not appear to benefit from adjuvant atezolizumab and the greatest benefit was observed in tumors with PD-L1 expression $\geq 50\%$. In the PEARLS trial, a DFS benefit was observed in the overall patient population, and not observed in the subset of patients with a tumor with TPS PD-L1 $\geq 50\%$. Whereas benefit was still seen in the neoadjuvant setting with nivolumab in Checkmate 816 in the overall study population and the subset of PD-L1 of patients with tumor PD-L1 $\geq 50\%$ [7, 35]. The validation of pCR or MPR in the immune therapy setting would also inform whether future clinical trials can utilize this endpoint—either to prove efficacy of a novel treatment strategy or to inform which patients should be selected for additional therapy in the adjuvant setting (i.e., those that did achieve a sufficient pathologic response following neoadjuvant therapy). Current investigations will focus on developing biomarkers of immunotherapy benefit in addition to PD-L1 level, and selecting patients at greater risk for relapse using minimal residual disease assays.

5 Adjuvant and Neoadjuvant Epidermal Growth Factor Receptor Tyrosine Kinase Inhibitor Trials

The initial trials of adjuvant EGFR TKIs were performed in patients who were not selected based on *EGFR* mutation status, but the next generation of trials investigated adjuvant EGFR TKIs in patients with sensitizing *EGFR* mutations [5, 6]. In the metastatic setting, first-generation EGFR TKIs had demonstrated superior objective response rates and longer progression-free survival compared with chemotherapy in patients with *EGFR* mutations. Several phase III trials compared first-generation EGFR TKIs with adjuvant cisplatin-based chemotherapy (Table 4) [36–39]. These trials assessed whether the standard therapy could be replaced with a better tolerated therapy and potentially more active therapy based on the data from metastatic disease. The primary endpoint was DFS, and the duration of EGFR TKIs was 2 years. All these trials, except the trial by Tada et al., demonstrated a statistically significant improvement in DFS (Table 4).

OS was a secondary endpoint in these trials, and they were not statistically powered to assess this endpoint. The interpretation of the OS is further complicated by the relatively short follow-up time, and a higher rate of cross over. The trials by Zhong et al. and Tada et al. did not reveal an improvement in OS on follow-up, and the OS data from the other trials are immature.

One observation from these trials is the rate of relapse increases after completion of EGFR TKI. The Kaplan–Meier curves for DFS of each treatment arm demonstrate separation during the time period patients are receiving EGFR TKI, and subsequently come together. This raises questions about the optimal duration of therapy. Despite the positive

Table 3 Select ongoing phase II trials evaluating adjuvant or neoadjuvant immune checkpoint inhibitors with chemotherapy

Trial ID ^a	Comparison	Stage	Primary outcome
NCT02486718 (IMpower010) [7]	Atezolizumab vs best supportive care following chemotherapy in patients with resected NSCLC	IB–IIIA	DFS
NCT02504372 (PEARLS) [8]	Pembrolizumab vs placebo after resection with or without standard adjuvant therapy	IB–IIIA	DFS
NCT02273375	Adjuvant durvalumab vs placebo in completely resected NSCLC	IB–IIIA	DFS
NCT02595944 (ANVIL)	Nivolumab vs standard observation after surgery and chemotherapy	IB–IIIA	DFS and OS
NCT03456063 (IMpower030)	Neoadjuvant atezolizumab with chemotherapy vs placebo with chemotherapy	IIA–IIIB	EFS
NCT04025879	Neoadjuvant and adjuvant nivolumab with chemotherapy vs placebo with chemotherapy	IIA–IIB	EFS
NCT03425643 (Keynote 671)	Neoadjuvant and adjuvant pembrolizumab with chemotherapy vs placebo with chemotherapy	II–IIIB	EFS and OS
NCT03800134 (AEGEAN)	Neoadjuvant and adjuvant durvalumab with chemotherapy vs placebo with chemotherapy	II–IIIB	EFS and pCR

DFS disease-free survival, EFS event-free survival, NSCLC non-small-cell lung cancer, OS overall survival, pCR pathologic complete response

^aClinicalTrials.gov identifier

Table 4 Randomized trials of adjuvant epidermal growth factor receptor tyrosine kinase inhibitors

Study	Phase	Comparison	Stage	DFS (median in months)	OS (median in months)
Zhong et al. [43]	III (n = 222)	Gefitinib Cisplatin/vinorelbine	II–IIIA	30.8 vs 19.8 HR 0.51, $p < 0.05$	75.5 vs 62.8 HR 0.92, $p = 0.67$
Tada et al. [37]	III (n = 234)	Gefitinib Cisplatin/vinorelbine	II–IIIA	35.9 vs 25.1 HR 0.92, $p = 0.63$	NR vs NR HR 1.03, $p = 0.89$
He et al. [38]	III (n = 322)	Icotinib Cisplatin-based therapy	II–IIIA	47.0 vs 22.2 HR 0.36, $p < 0.05$	NR vs NR HR 0.91 ^a
Yue et al. [39]	II (n = 102)	Erlotinib Cisplatin/vinorelbine	IIIA	42.4 vs 21.0 HR 0.268, $p < 0.05$	NR vs NR HR 0.165 ^a
Wu et al. [41]	III (n = 682)	Osimertinib Placebo	IB–IIIA	NR vs 27.5 HR 0.20 $p < 0.05$	NR vs NR HR 0.40 ^a

DFS disease-free survival, HR hazard ratio, NR not reached, OS overall survival

^aData are immature

DFS observed in these trials, the first generation EGFR TKIs have not been widely adopted as standard-of-care adjuvant therapies.

The third-generation TKI osimertinib had revealed superior efficacy to first-generation EGFR TKIs in patients with metastatic *EGFR* mutant NSCLC [40]. This agent was investigated in a phase III trial of osimertinib compared with placebo in resected *EGFR* mutant NSCLC, and revealed a statistically significant improvement in DFS [41]. The primary analysis was performed in patients with resected stage II or III NSCLC, and revealed an improvement in DFS (HR 0.17, 95% CI 0.11–0.26; $p < 0.001$), and the overall patient population, encompassing stage IB–IIIA patients, revealed an improvement in DFS as well (HR 0.20, 95% CI 0.14–0.30; $p < 0.001$). The benefit was similar in patients who did or did not receive adjuvant chemotherapy. Central nervous system (CNS) recurrence can be a significant clinical issue and CNS recurrence favored osimertinib (HR 0.18, 95% CI 0.10–0.33). The data are immature for the secondary endpoint of OS, but the dramatic DFS improvement, and meaningful delay in CNS recurrence, have prompted many to view osimertinib as a standard-of-care adjuvant therapy. In line with this rationale, the US FDA and European Medicines Agency (EMA) approved adjuvant osimertinib in patients with resected stage IB–IIIA NSCLC in patients with a sensitizing *EGFR* mutation.

Several single-arm, phase II trials have investigated neoadjuvant EGFR TKIs (erlotinib or gefitinib) in patients with *EGFR* mutant NSCLC. A pooled analysis of five prospective trials revealed an ORR of 58.5%, and in the subset of patients with stage IIIA disease, an ORR of 51.4% was observed [42]. Data was available on pCR from two trials, and the rate observed was 0%. A randomized controlled trial compared neoadjuvant erlotinib with cisplatin and gemcitabine [43]. The primary endpoint was ORR; the ORR in the erlotinib and cisplatin/gemcitabine arms was 54.1% and 34.3%, respectively (odds ratio 2.26, 95% CI 0.87–5.84; $p = 0.092$).

No pCR were observed in either arm, and in the erlotinib and cisplatin/gemcitabine arms MPR was observed in 9.7% and 0%, respectively. The secondary endpoint of PFS was longer in the erlotinib arm; however, patients received 12 months of adjuvant erlotinib. It cannot be determined if the longer PFS was related to the neoadjuvant and/or adjuvant erlotinib.

A phase III trial with three arms is currently evaluating osimertinib monotherapy, osimertinib plus chemotherapy, or chemotherapy plus placebo as neoadjuvant therapy for early-stage NSCLC (NCT04351555).

6 Additional Targeted Therapies

The development of other targeted therapies as neoadjuvant and adjuvant therapy has been slower than in *EGFR* mutant NSCLC due to the lower prevalence. Several case reports and a case series describe the use of pre-operative anaplastic lymphoma kinase (ALK) tyrosine kinase inhibitors in patients with ALK + NSCLC [44–47]. In the case series, ten of the eleven patients had a partial response and two patients had a pCR. Given the small size of these reports, pre-operative ALK TKIs cannot be considered standard of care. Several studies investigating adjuvant ALK TKIs are ongoing in patients with surgically resected ALK + NSCLC, including a phase III trial comparing alectinib with adjuvant platinum-based therapy in patients with resected stage NSCLC (NCT03456076), and a phase III trial investigating crizotinib compared with observation in patients with resected NSCLC after chemotherapy (NCT02201992) [9]. The role of adjuvant ALK TKIs will be defined by the results of these trials.

Given the difficulty in identifying and enrolling patients with molecular alterations to neoadjuvant trials, an ongoing trial will use centralized plasma and tumor genotyping, and patients will be assigned to a specific trial investigating neoadjuvant targeted therapy in a specific molecular

subset (NCT04351555, NCT04302025, NCT04759911, NCT04926831). This approach of central testing and investigating multiple targeted therapies as part of a master protocol may accelerate the development of targeted therapies in surgically resected NSCLC, particularly for less common but targetable genomic alterations (*ROS1* rearrangements, *MET* exon 14 alterations, *RET* rearrangements, and *BRAF* V600E mutations).

7 Checkpoint Inhibitors Versus Targeted Therapy

In the metastatic disease setting, immune checkpoint inhibitors have been less effective for patients with driver alteration-positive NSCLC (*EGFR* mutations, *ALK* rearrangements, and *ROS1* rearrangements) [48]. In the early-stage setting, a subset analysis of the IMpower 010 found that patients with an *EGFR* mutation, stage II–IIIA disease, and a PD-L1 $\geq 1\%$ ($n = 43$) appeared to favor atezolizumab (HR 0.57, 95% CI 0.26–1.24) [7]. However, the small sample size of this subgroup limits interpretation. The robust and statistically significant DFS seen in the ADURA trial, along with experience from the metastatic setting, has led the current clinical practice to use adjuvant osimertinib rather than atezolizumab in patients with an *EGFR* mutation. Notably, in the subset of patients with ALK+, PD-L1 + NSCLC in IMPower010 ($n = 23$), the DFS with atezolizumab compared with best supportive care was similar (HR 1.05, 95% CI 0.32–3.45). Thus, although the number of patients with known driver alterations treated with adjuvant checkpoint inhibitors in trials remains small, the benefit of adjuvant immune checkpoint inhibitors appears modest in this population. Several ongoing phase III trials evaluating immune checkpoint inhibitors in the neoadjuvant or adjuvant setting do not exclude *EGFR* or *ALK* alteration-positive NSCLC (NCT02504372, NCT04025879, NCT03425643) [9]. However, genomic testing is not universal in the early-stage setting and for many patients in these trials the genomic alteration status remains unknown [7].

8 Conclusions

The past decade has seen a substantial improvement in the therapeutic treatment landscape for NSCLC with the use of molecularly targeted therapies and immune checkpoint inhibitors. These advancements are now extended to the early-stage disease setting with impressive disease-free survival benefit and the subsequent FDA and European Medical Agency approval of osimertinib as adjuvant therapy for *EGFR* mutated NSCLC, and atezolizumab adjuvant therapy for patients with the required level of PD-L1 expression.

Longer term follow up is needed to determine whether these therapies merely delay recurrence or can improve overall survival and increase the cure rate following surgical resection. Future research efforts will focus on applying biomarkers such as PD-L1, genomic sequencing, or molecular residual disease to prioritize patient groups most likely to benefit from checkpoint inhibitors or target therapy in the early-stage setting. Novel endpoints such as pathologic response at the time of surgery following neoadjuvant therapy may accelerate the development and validation of these treatment regimens. Ultimately, these efforts aim to personalize therapy and improve outcomes in the early-stage disease setting when the cancer burden is low and long-term disease control or even cure are possible.

Declarations

External funding No external funding was used in the preparation of this manuscript.

Conflicts of interest/competing interests James Isaacs received salary support from Duke University of North Carolina Immunotherapy T32 Training Program, and has received honoraria from MJH Life Sciences Participation in Clinical Congress Consultants, and ASCO panel review for oncology fellows. Thomas E. Stinchcombe has received honorarium from participation in advisory boards from Takeda, AstraZeneca, Genentech/Roche, Foundation Medicine, Pfizer, EMD Serono, Novartis, Daiichi Sankyo, Lilly, Medtronic, Puma Biotechnology, Janssen Oncology, Regeneron, Turning Point Therapeutics, Sanofi/Aventis, and has received grant support for clinical trials from Genentech/Roche, AstraZeneca, Takeda, Advaxis, Regeneron, and Mirati.

Ethics approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

Availability of data and material Not applicable.

Code availability Not applicable.

Author contributions JI contributed to the conceptualization, investigation, and writing (original draft, review and editing) and TS contributed to the conceptualization, investigation, and writing (original draft, review and editing).

References

1. Goldstraw P, Chansky K, Crowley J, Rami-Porta R, Asamura H, Eberhardt WEE, et al. The IASLC lung cancer staging project: proposals for revision of the TNM stage groupings in the forthcoming (eighth) edition of the TNM classification for lung cancer. *J Thorac Oncol*. 2016;11:39–51.
2. Pignon JP, Tribodet H, Scagliotti GV, Douillard JY, Shepherd FA, Stephens RJ, et al. Lung adjuvant cisplatin evaluation: a pooled analysis by the LACE Collaborative Group. *J Clin Oncol*. 2008;26:3552–9.

3. Song WA, Zhou NK, Wang W, Chu XY, Liang CY, Tian XD, et al. Survival benefit of neoadjuvant chemotherapy in non-small cell lung cancer: an updated meta-analysis of 13 randomized control trials. *J Thorac Oncol.* 2010;5:510–6.
4. Wakelee HA, Dahlberg SE, Keller SM, Tester WJ, Gandara DR, Graziano SL, et al. Adjuvant chemotherapy with or without bevacizumab in patients with resected non-small-cell lung cancer (E1505): an open-label, multicentre, randomised, phase 3 trial. *Lancet Oncol.* 2017;18:1610–23.
5. Goss GD, O'Callaghan C, Lorimer I, Tsao M-S, Masters GA, Jett J, et al. Gefitinib versus placebo in completely resected non-small-cell lung cancer: results of the NCIC CTG BR19 study. *J Clin Oncol.* 2013;31:3320–6.
6. Kelly K, Altorki NK, Eberhardt WE, O'Brien ME, Spigel DR, Crinò L, et al. Adjuvant erlotinib versus placebo in patients with stage IB–IIIA non-small-cell lung cancer (RADIANT): a randomized, double-blind, phase III trial. *J Clin Oncol.* 2015;33:4007–14.
7. Felip E, Altorki N, Zhou C, Csőszi T, Vynnychenko I, Goloborodko O, et al. Adjuvant atezolizumab after adjuvant chemotherapy in resected stage IB–IIIA non-small-cell lung cancer (IMpower010): a randomised, multicentre, open-label, phase 3 trial. *Lancet.* 2021;398:1344–57.
8. Paz-Ares L, O'Brien MER, Mauer M, Dafni U, Oselin K, Havel L, et al. VP3-2022: pembrolizumab (pembro) versus placebo for early-stage non-small cell lung cancer (NSCLC) following complete resection and adjuvant chemotherapy (chemo) when indicated: randomized, triple-blind, phase III EORTC-1416-LCG/ETOP 8–15 & #x2013; PEARLS/KEYNOTE-091 study. *Ann Oncol.* 2022;33:451–3.
9. Clinicaltrials.gov. Accessed 3 Jan 2022.
10. Uprety D, Mandrekar SJ, Wigle D, Roden AC, Adjei AA. Neoadjuvant immunotherapy for NSCLC: current concepts and future approaches. *J Thorac Oncol.* 2020;15:1281–97.
11. Blumenthal GM, Bunn PA Jr, Chaft JE, McCoach CE, Perez EA, Scagliotti GV, et al. Current status and future perspectives on neoadjuvant therapy in lung cancer. *J Thorac Oncol.* 2018;13:1818–31.
12. Topalian SL, Taube JM, Pardoll DM. Neoadjuvant checkpoint blockade for cancer immunotherapy. *Science.* 2020;367:6477.
13. Pisters KM, Kris MG, Gralla RJ, Zaman MB, Heelan RT, Martini N. Pathologic complete response in advanced non-small-cell lung cancer following preoperative chemotherapy: implications for the design of future non-small-cell lung cancer combined modality trials. *J Clin Oncol.* 1993;11:1757–62.
14. Mouillet G, Monnet E, Milleron B, Puyraveau M, Quoix E, David P, et al. Pathologic complete response to preoperative chemotherapy predicts cure in early-stage non-small-cell lung cancer: combined analysis of two IFCT randomized trials. *J Thorac Oncol.* 2012;7:841–9.
15. Hellmann MD, Chaft JE, William WN Jr, Rusch V, Pisters KM, Kalhor N, et al. Pathological response after neoadjuvant chemotherapy in resectable non-small-cell lung cancers: proposal for the use of major pathological response as a surrogate endpoint. *Lancet Oncol.* 2014;15:e42–50.
16. Junker K, Thomas M, Schulmann K, Klink F, Bosse U, Müller KM. Tumour regression in non-small-cell lung cancer following neoadjuvant therapy. Histological assessment. *J Cancer Res Clin Oncol.* 1997;123:469–77.
17. Travis WD, Dacic S, Wistuba I, Sholl L, Adusumilli P, Bubendorf L, et al. IASLC multidisciplinary recommendations for pathologic assessment of lung cancer resection specimens after neoadjuvant therapy. *J Thorac Oncol.* 2020;15:709–40.
18. Pataer A, Kalhor N, Correa AM, Raso MG, Erasmus JJ, Kim ES, et al. Histopathologic response criteria predict survival of patients with resected lung cancer after neoadjuvant chemotherapy. *J Thorac Oncol.* 2012;7:825–32.
19. Chaft JE, Rusch V, Ginsberg MS, Paik PK, Finley DJ, Kris MG, et al. Phase II trial of neoadjuvant bevacizumab plus chemotherapy and adjuvant bevacizumab in patients with resectable nonsquamous non-small-cell lung cancers. *J Thorac Oncol.* 2013;8:1084–90.
20. Qu Y, Emoto K, Eguchi T, Aly RG, Zheng H, Chaft JE, et al. Pathologic assessment after neoadjuvant chemotherapy for NSCLC: importance and implications of distinguishing adenocarcinoma from squamous cell carcinoma. *J Thorac Oncol.* 2019;14:482–93.
21. Cottrell TR, Thompson ED, Forde PM, Stein JE, Duffield AS, Anagnostou V, et al. Pathologic features of response to neoadjuvant anti-PD-1 in resected non-small-cell lung carcinoma: a proposal for quantitative immune-related pathologic response criteria (irPRC). *Ann Oncol.* 2018;29:1853–60.
22. Stein JE, Lipson EJ, Cottrell TR, Forde PM, Anders RA, Cimino-Mathews A, et al. Pan-tumor pathologic scoring of response to PD-(L)1 blockade. *Clin Cancer Res.* 2020;26:545–51.
23. Liu J, Blake SJ, Yong MC, Harjunpää H, Ngiew SF, Takeda K, et al. Improved efficacy of neoadjuvant compared to adjuvant immunotherapy to eradicate metastatic disease. *Cancer Discov.* 2016;6:1382–99.
24. Blank CU, Rozeman EA, Fanchi LF, Sikorska K, van de Wiel B, Kvistborg P, et al. Neoadjuvant versus adjuvant ipilimumab plus nivolumab in macroscopic stage III melanoma. *Nat Med.* 2018;24:1655–61.
25. Forde PM, Chaft JE, Smith KN, Anagnostou V, Cottrell TR, Hellmann MD, et al. Neoadjuvant PD-1 blockade in resectable lung cancer. *N Engl J Med.* 2018;378:1976–86.
26. Lee JM, Chaft J, Nicholas A et al. Surgical and clinical outcomes with neoadjuvant atezolizumab in resectable stage IB–IIIB NSCLC: LCMC3 trial primary analysis. In: 2020 World conference on lung cancer abstract PS0205 2021.
27. Cascone T, William WN, Weissferdt A, Leung CH, Lin HY, Pataer A, et al. Neoadjuvant nivolumab or nivolumab plus ipilimumab in operable non-small cell lung cancer: the phase 2 randomized NEOSTAR trial. *Nat Med.* 2021;27:504–14.
28. Gao S, Li N, Gao S, Xue Q, Ying J, Wang S, et al. Neoadjuvant PD-1 inhibitor (Sintilimab) in NSCLC. *J Thorac Oncol.* 2020;15:816–26.
29. Wislez MMJ, Lavole A, Zalcman G, Carre O, Egenod T, Caliendo R, Gervais R. Neoadjuvant durvalumab in resectable non-small cell lung cancer (NSCLC): preliminary results form a multicenter study (IFCT-1601 IONESCO). *Ann Oncol.* 2020;31(suppl_4):S794.
30. Besse B, Adam J, Cozic N, Chaput-Gras N, Planchard D, Mezquita L, et al. 1215O-SC Neoadjuvant atezolizumab (A) for resectable non-small cell lung cancer (NSCLC): results from the phase II PRINCEPS trial. *Ann Oncol.* 2020;31:S794–5.
31. Reuss JE, Anagnostou V, Cottrell TR, Smith KN, Verde F, Zahurak M, et al. Neoadjuvant nivolumab plus ipilimumab in resectable non-small cell lung cancer. *J Immunother Cancer.* 2020;8: e001282.
32. Shu CA, Gainor JF, Awad MM, Chiuhan C, Grigg CM, Pabani A, et al. Neoadjuvant atezolizumab and chemotherapy in patients with resectable non-small-cell lung cancer: an open-label, multicentre, single-arm, phase 2 trial. *Lancet Oncol.* 2020;21:786–95.
33. Provencio M, Nadal E, Insa A, García-Campelo MR, Casal-Rubio J, Dómine M, et al. Neoadjuvant chemotherapy and nivolumab in resectable non-small-cell lung cancer (NADIM): an open-label, multicentre, single-arm, phase 2 trial. *Lancet Oncol.* 2020;21:1413–22.
34. Rothschild SI, Zippelius A, Eboulet EI, Prince SS, Betticher D, Bettini A, et al. SAKK 16/14: durvalumab in addition to neoadjuvant chemotherapy in patients with stage IIIA(N2) non-small-cell

- lung cancer—a multicenter single-arm phase II trial. *J Clin Oncol*. 2021;39:2872–80.
35. Forde PM, Spicer J, Lu S, Provencio M, Mitsudomi T, Awad MM et al. Neoadjuvant nivolumab plus chemotherapy in resectable lung cancer. *N Engl J Med*. 2022.<https://doi.org/10.1056/NEJMo a2202170>
 36. Zhong WZ, Wang Q, Mao WM, Xu ST, Wu L, Wei YC, et al. Gefitinib versus vinorelbine plus cisplatin as adjuvant treatment for stage II–IIIA (N1–N2) EGFR-mutant NSCLC: final overall survival analysis of CTONG1104 phase III trial. *J Clin Oncol*. 2021;39:713–22.
 37. Tada H, Mitsudomi T, Misumi T, Sugio K, Tsuboi M, Okamoto I, et al. Randomized phase III study of gefitinib versus cisplatin plus vinorelbine for patients with resected stage II–IIIA Non-small-cell lung cancer with EGFR mutation (IMPACT). *J Clin Oncol*. 2022;40:231–41.
 38. He J, Su C, Liang W, Xu S, Wu L, Fu X, et al. Icotinib versus chemotherapy as adjuvant treatment for stage II–IIIA *EGFR*-mutant non-small-cell lung cancer (EVIDENCE): a randomised, open-label, phase 3 trial. *Lancet Respir Med*. 2021;9:1021–9.
 39. Yue D, Xu S, Wang Q, Li X, Shen Y, Zhao H, et al. Erlotinib versus vinorelbine plus cisplatin as adjuvant therapy in Chinese patients with stage IIIA *EGFR* mutation-positive non-small-cell lung cancer (EVAN): a randomised, open-label, phase 2 trial. *Lancet Respir Med*. 2018;6:863–73.
 40. Ramalingam SS, Vansteenkiste J, Planchard D, Cho BC, Gray JE, Ohe Y, et al. Overall survival with osimertinib in untreated, *EGFR*-mutated advanced NSCLC. *N Engl J Med*. 2019;382:41–50.
 41. Wu Y-L, Tsuboi M, He J, John T, Grohe C, Majem M, et al. Osimertinib in resected *EGFR*-mutated non-small-cell lung cancer. *N Engl J Med*. 2020;383:1711–23.
 42. Sun L, Guo Y-J, Song J, Wang Y-R, Zhang S-L, Huang L-T, et al. Neoadjuvant *EGFR*-TKI therapy for *EGFR*-mutant NSCLC: a systematic review and pooled analysis of five prospective clinical trials. *Front Oncol (Syst Rev)*. 2021;10:3021.
 43. Zhong WZ, Chen KN, Chen C, Gu CD, Wang J, Yang XN, et al. Erlotinib versus gemcitabine plus cisplatin as neoadjuvant treatment of stage IIIA–N2 *EGFR*-mutant non-small-cell lung cancer (EMERGING-CTONG 1103): a randomized phase II study. *J Clin Oncol*. 2019;37:2235–45.
 44. Imanishi N, Yoneda K, Taira A, Ichiki Y, Sato N, Hisaoka M, et al. Major pathologic response to alectinib in *ALK*-rearranged adenocarcinoma of the lung. *Surg Case Rep*. 2018;4:19.
 45. Tian Y, Huang J, Li C, Jiang L, Lin H, Lu P, et al. Perioperative crizotinib in a patient with stage IIIB *ALK*-positive non-small cell lung cancer: a case report. *Ann Transl Med*. 2020;8:770.
 46. Zhang C, Li SL, Nie Q, Dong S, Shao Y, Yang XN, et al. Neoadjuvant crizotinib in resectable locally advanced non-small cell lung cancer with *ALK* rearrangement. *J Thorac Oncol*. 2019;14:726–31.
 47. Xie XH, Zhan ZJ, Qin YY, Jiang JH, Yin WQ, Zheng RH, et al. Case report: neoadjuvant and adjuvant crizotinib targeted therapy in stage IIIA–N2 *ALK*-positive non-small-cell lung cancer. *Front Oncol*. 2021;11: 655856.
 48. Mazieres J, Drilon A, Lusque A, Mhanna L, Cortot AB, Mezquita L, et al. Immune checkpoint inhibitors for patients with advanced lung cancer and oncogenic driver alterations: results from the IMMUNOTARGET registry. *Ann Oncol*. 2019;30:1321–8.